CITY OF CAMBRIDGE
POLICE DEPARTMENT

125 Sixth Street. Cambridge, Massachusetts 02142 (617) 349-3300 www.cambridgepolice.org

SOUNDTHINKING™
for greater public safety™

ShotSpotter®

Cambridge Police ShotSpotter Program

What is ShotSpotter:

- ShotSpotter is a network of acoustic sensors that can detect, locate, and alert police to gunshot incidents.
- Installation of the system was approved by the Cambridge City Council in 2014.
- All costs associated with the system for Cambridge and several other cities are paid for as part of an Urban Area Security Initiative (UASI) grant program.
- SoundThinking, Inc., the owner of ShotSpotter, installed the sensors and continues to maintain them.
- Locations of the sensors is not publicly disclosed to protect the integrity of the system.

ShotSpotter®

SOUNDTHINKING™
for greater public safety™

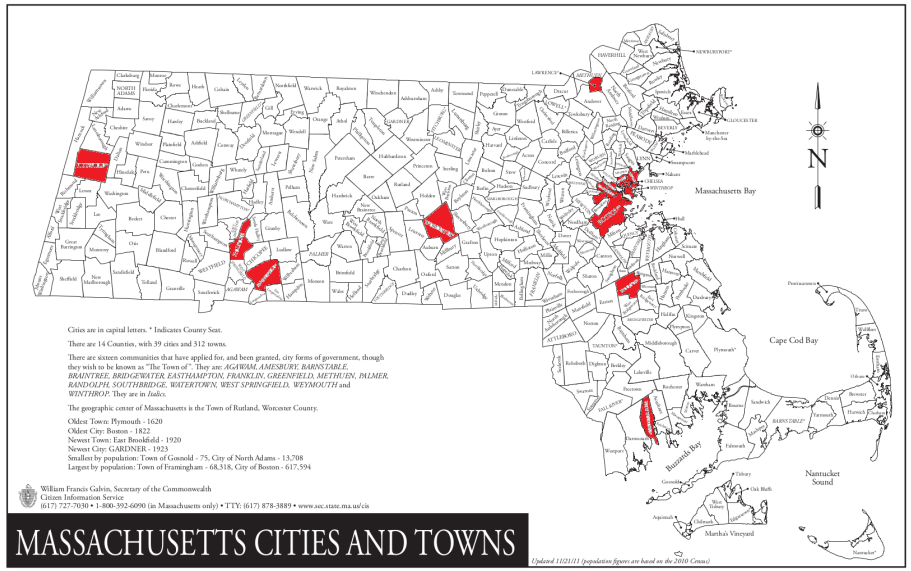
Who Uses ShotSpotter in Ma:

Greater Boston UASI Partners:

1. Boston
2. Cambridge
3. Chelsea
4. Everett
5. Revere
6. Somerville

Other Massachusetts Users:

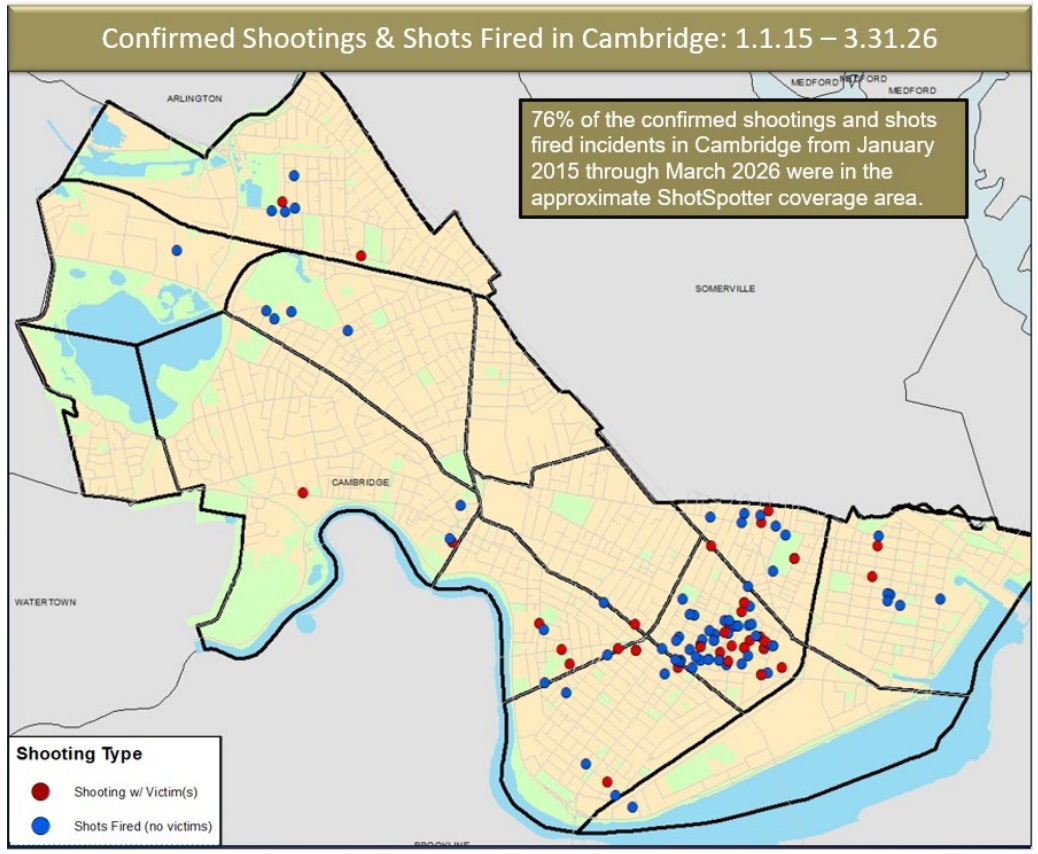
1. Brockton
2. Holyoke
3. Lawrence
4. New Bedford
5. Pittsfield
6. Springfield
7. Worcester
8. Northeastern University



**Highlighted cities have passed resolutions declaring themselves as "welcoming communities"*

Coverage Area:

- Coverage area has been determined using historical data on shooting incidents from the CPD Crime Analysis.
- In the 10+ years the system has been operational, 76% of confirmed shootings city-wide have taken place in the coverage area.
- The current coverage area is approximately 1.25 square miles and covers several neighborhoods in the city.
- Any expansion of the system would be done through a public process with input from community stakeholders.



Breaking it Down:

For ShotSpotter to activate there needs to be an explosive noise or sound measuring 120 decibels or greater that hits 3 or more sensors.

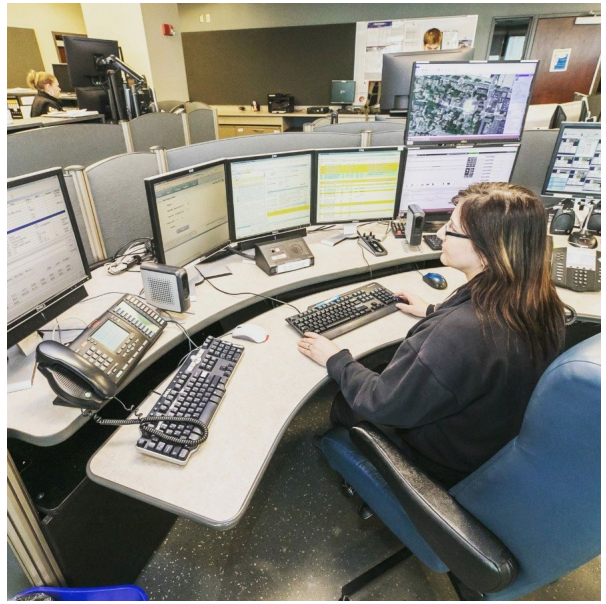
- **For comparison, human voice averages 60 – 65 decibels**

2018 – Present in Cambridge:

- **Number of alerts:** **129**
- **Rounds of Gunfire (shots) Detected:** **418**

9-1-1 Call vs. ShotSpotter:

- ShotSpotter is an automated technology that is activated by potential gunshot incidents and verified in real time by trained professionals. This results in prompt notifications to law enforcement, which can facilitate a faster response.
- Traditional 911 calls are dependent on people taking action. If a person does not have immediate access to a phone, know their location or chooses not to call, it adversely impacts the emergency response.
- From 2015 – Present there have been 11 confirmed shootings that were picked up by ShotSpotter but did not result in any 9-1-1 calls.



9-1-1 Call vs. ShotSpotter:

Percentage of 9-1-1 calls confirmed to be incidents of gunfire in Cambridge:

19%

ShotSpotter Activations in Cambridge that have been confirmed to be incidents of gunfire:

35%

- Data analyzed by the Cambridge Police Crime Analysis Unit over the last decade has shown that in Cambridge, ShotSpotter reports of gunfire are more often verified than those reported through the 9-1-1 system.
- The ShotSpotter system will typically provide a much smaller location radius than a 9-1-1 call.

Medical Aid:

Officers get ShotSpotter alerts directly on their phones and cruiser computers, which has led to a faster response and prompt medical assistance, when seconds can make the difference:

May 23, 2024: Donnelly Field Shooting:

"A tourniquet was applied by an officer while other officers assisted with applying pressure to the wound. A trauma wound dressing was also applied shortly after. At this time, Cambridge Fire and Pro EMS arrived"

- Officer Incident Report

July 16, 2024: Harvard Street Shooting:

"I was unable to locate any wound due to the amount of blood covering the victims legs so I applied a CAT tourniquet as high as possible. After the tourniquet was applied, one bullet entry wound was located high on the victims inner thigh. Due to the wounds location, officers began applying hemostatic gauze until he was unable to pack the wound anymore. I the applied a trauma wound dressing on top of the wound. I then assisted Cambridge Fire with loading the victim onto a stretcher."

- Officer Incident Report

No 9-1-1 Call:

“was laying in his bedroom when he heard 3 loud bangs that he initially did not believe to be gunshots which is why he did not call 911”

A neighbor spoke with officers at the scene of a July 21, 2024 gun shot incident on Fairmont St. located through ShotSpotter

9-1-1 Call and Witness Statements:

Witness accounts in gun shot cases are often vague, at times conflicting and often cannot provide an accurate location. Here are the 5 witness statements from the July 28, 2024 gun shot incident on Putnam Ave. As you can see the number of shots heard and the location vary:

1. *"Heard 5 – 6 gunshots", "Did not know exactly where they came from"*
2. *"She heard 7 gunshots", Believes she heard the gunshots coming from the area of Putnam Ave and River Street"*
3. *"Heard 8 gunshots come from the area of Hoyt Field"*
4. *"They heard 1 gunshot, a pause, then 5 more", "They were not sure exactly where the shots came from, but stated they were very loud."*
5. *"They heard at least 5 gunshots. They were unsure where exactly the gunshots came from."*

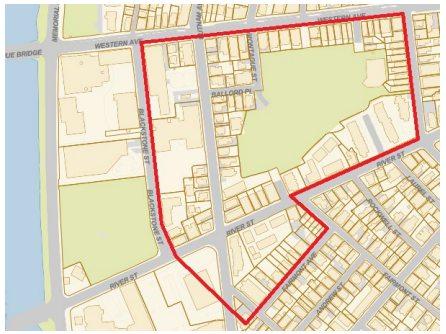
Compared to ShotSpotter:

Witness Round Count:

- Five witnesses, range between 5 and 8 gunshots.

Witness Location:

- Three witnesses not sure of the location.
- One witness said the vicinity of Hoyt Field.
- One witness said the intersection of Putnam and River St.





ShotSpotter Round Count:

- Audio recording of 8 rounds.

ShotSpotter Location:

- Parking lot between Putnam Ave and Blackstone St. opposite Riverside Press Park.

How Cambridge Responds:

- The initial number of officers dispatched to a 9-1-1 call reporting gunshots and a ShotSpotter activation is the same.
- With a ShotSpotter activation, Officers working will receive the alert directly on their phone and cruiser computer.
- When Emergency Communications (ECC) receives an alert and will immediately dispatch nearby officers to investigate.
- When officers arrive they look for any active threat, victims, ballistic evidence or potential witnesses.
  - If it is confirmed that a shooting occurred additional resources will respond to assist victims, process the crime scene and speak with witnesses.
  - If there is no evidence of a shooting, officers will clear the area and return to normal patrol duties.



Unfounded Alerts:

An unfounded alert does not definitively mean a shooting did not occur.

- More simply put, an unfounded alert means no evidence was located to prove there had been a shooting.
  - No victims
  - No ballistic evidence
  - No witnesses

- Occasionally alerts are proven to be false, mainly caused due to the following:
  - Loud industrial work or construction noises
  - Fireworks
  - A car backfiring or loudly crashing
  - Electrical transformer explosion.

Common Misconceptions:

Does ShotSpotter record private conversations? - No

The system is only activated when 3 or more sensors are triggered by an explosive sound measuring 120 decibels. For comparison, an average human voice is 60 decibels and will not trigger any sensors. **There has never been a conversation recorded in Cambridge**

Does ShotSpotter result in over-policing? - No

Massachusetts Courts have said a ShotSpotter alert alone does not give a police officer reasonable suspicion to stop someone. **There have never been any use of force incidents or community complaints resulting from a CPD response to a ShotSpotter alert.**

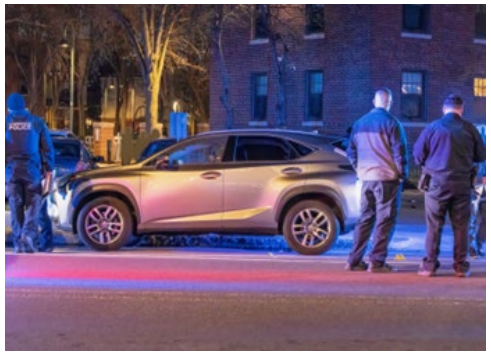
Is ShotSpotter Utilized for Immigration Enforcement? - No

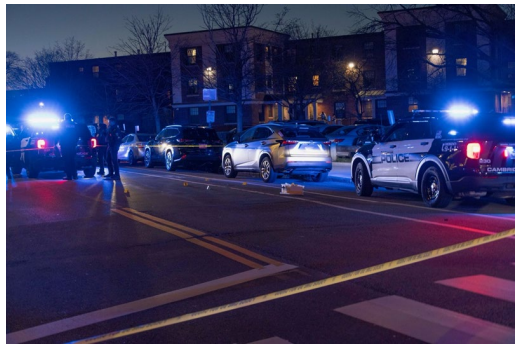
Data captured by ShotSpotter (explosive sounds registering 120 decibels) would not have any use to immigration enforcement. **There have been no incidents where Immigration & Customs Enforcement (ICE) has requested any ShotSpotter data from CPD or SoundThinking**. Release of any such data would require a judicial subpoena and be limited to a specific criminal investigation of a gunshot incident.

Cambridge Shooting Stats:

2024 Calendar Year:

- Total number of confirmed shooting incidents citywide - **14**
- Total shooting incidents within the ShotSpotter coverage area - **10**
- Two instances where a ShotSpotter alert resulted in a faster police response than waiting for 9-1-1 calls.





2025 Calendar Year:

- Total number of confirmed shooting incidents citywide - **9**
- Total shooting incidents within the ShotSpotter coverage area - **7**



Real World Example:

- On August 27, 2022 at 12:34am police responded to a ShotSpotter Activation (audio attached) for two gunshots in the vicinity of Putnam Gardens. There were no 9-1-1 calls, only a Citizen Observer tip at 12:48am stating they'd heard 2 shots near Magee St.
- Officers arrived on-scene within 2 minutes of the ShotSpotter activation. The suspect had fled, but it is unclear if they were deterred by the police response.
- Two spent shell casings were discovered which led detectives to speak with neighbors and review video footage from the vicinity.
- It was learned that this was related to a high-risk domestic violence case.
- The suspect was identified and criminally charged in Middlesex Superior Court, where he was found guilty of multiple charges.



Real World Example:

- On September 8, 2022 at 4:07am police responded to a ShotSpotter Activation for a single gunshot in the middle of Donnelly Field.
- There were no 9-1-1 calls received regarding this incident. Upon arrival officers encountered several individuals in the park who provided vague statements that attributed the activation to fireworks.
- Officers searched the park and ultimately recovered a 9MM Beretta Handgun with 14 rounds of ammunition that had been discarded in the grass.
- Given that the incident occurred in the early morning hours and in close proximity to a school it is concerning to think what may have happened if officers had not responded and recovered the weapon.



Safeguarding Our Community:

Impact on Police Response:

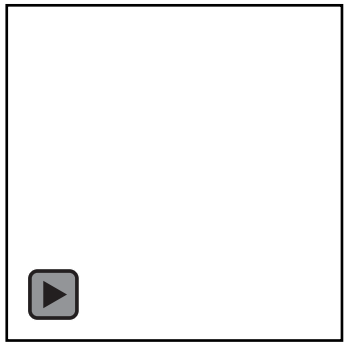
- Use of Force resulting from ShotSpotter Activations: 0
- Complaints resulting from response to ShotSpotter Activations: 0

Protecting Our Information:

- Only select Cambridge Police Department personnel have access to our ShotSpotter Data.
- CPD controls all sharing and distribution of alerts or other material collected.
- Currently, the only thing shared outside the Cambridge Police Department is alert notifications and case materials:
  - Alert notifications are shared with select members of the Somerville Police Department to aid in emergency response and investigations as history has shown a strong correlation between gun violence in both communities.
  - Case materials are shared with members of the Middlesex District Attorney's Office for their use during active criminal investigations and criminal prosecutions.

Benefits to Cambridge:

- ✓ Can alert officers to gunshots faster than a traditional 9-1-1 call, resulting in a quicker emergency response that has the potential to save lives.
- ✓ Provides first responders with a more precise location of a potential incident than traditional 9-1-1 callers are typically able to.
- ✓ Makes police aware of an incident even when there aren't any 9-1-1 calls.
- ✓ Helps investigators better understand what took place in a confirmed shooting incident.





QUESTIONS, COMMENTS, & DISCUSSION: